

DAVID BRAUN, Ph.D.

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PROFESSIONAL SUMMARY

Quantitative analyst and model developer with PhD + 5 years postdoc experience building and validating predictive models of complex human behavior, including decision-making, attention, and risk preferences. Skilled in supervised learning, classification (tree-based models, logistic regression), and object-oriented programming, with a growing focus on model interpretability and real-world deployment. Proven ability to translate ambiguous research questions into high-performance, data-driven solutions. Passionate about applying machine learning to socially impactful domains like fraud prevention and financial integrity.

SKILLS

Programming: R (10 yrs, *tidyverse*, *ggplot*), Python (9 yrs; *scikit-learn*, *PyTorch*), Git/GitHub, Stan, SQL

Statistical & Machine Learning: Linear / logistic / hierarchical / ridge regression, random forest, clustering, PCA, deep learning, predictive modeling, hypothesis testing, hierarchical Bayesian computational modeling

Collaboration: Co-led and published research with cognitive scientists, neuroscientists, epidemiologists, computer scientists, material scientists, engineers, and economists

Communication: Published 10+ papers in scientific, peer-reviewed journals. Presented research orally at 25+ conferences and colloquia to technical and non-technical audiences

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | Neuroscience, [Drexel University](#), Philadelphia, PA

Oct 2023 – Present

- Leveraged **unsupervised machine learning** (cluster-based permutation testing) to identify biomarkers of anxiety-related interoceptive dysfunction in 174GB of multimodal EEG/ECG ([first-authored paper](#); [poster](#); [blog post](#)).
- Applied **novel individual-level PCA and hierarchical clustering method** to uncover unique patterns in spontaneous thinking, advancing personalized diagnostics in mental health and precision medicine ([GitHub](#); [poster](#)).
- Independently developed a Bayesian model** investigating neural and cognitive mechanisms underlying adaptive vs. maladaptive forms of off-task thinking for NIH early-career grant proposal (see [Specific Aims](#)).
- Built and maintained automated Python ETL pipeline** (1,500+ LOC) to convert raw EEG, fMRI, and behavioral data from multiple lab experiments into an industry-standardized format, significantly enhancing reproducibility and cutting manual processing time ([GitHub](#)).

Postdoctoral Researcher | Cognition, [Lehigh University](#), Bethlehem, PA

Sept 2020 – Aug 2023

- Led evaluation of COVID-19 non-pharmaceutical interventions (NPIs) **using Bayesian timeseries modeling, random forest** and 10,000+ crowdsourced responses, providing evidence that perceived public compliance with NPIs can improve forecasts 3 weeks ahead ([first-authored paper](#); [data/code on OSF](#); [poster](#)).
- Built an **object detection model using transfer learning** to automate labeling of eye-tracking data in an observational study comparing expert vs. novice performance across traditional and novel apparatuses ([paper](#)).
- Validated a public concern scale for cyber-physical technologies ($N > 1,000$) using **factor analysis and SEM**; developed a **computational model of public concern**, aiding engineers' simulations in an NSF-funded project.
- Mentored** 3 undergraduates in independent research projects ([example student poster](#)).

Graduate Researcher, [Lehigh University](#), Bethlehem, PA

Sept 2014 – May 2020

- Developed hierarchical logistic models to quantify individual-level risk preferences** in effort-based decision contexts, uncovering systematic variation in risk tolerance across real and hypothetical scenarios, with implications for behavioral modeling and decision support ([example analysis](#)).
- Published a novel paradigm quantifying effort-based **decision making** ([first-authored paper](#); 35+ citations; [poster](#)).
- Taught R programming** as lead instructor in both undergraduate and graduate courses / seminars, fostering technical proficiency and data literacy across diverse skill levels ([graduate syllabus](#)).

EDUCATION

Ph.D. Cognitive Psychology | Lehigh University | May 2020

M.S. Cognitive Psychology | Lehigh University | Dec 2016

B.A. Psychology | Stockton University | May 2013